Haoyang Zhang

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Research Focus

I build efficient systems for AI on emerging hardware such as CPU–GPU superchips, co-designing LLM serving and training (Mixture-of-Experts inference, speculative decoding) with the memory hierarchy, interconnects, and GPU kernels. My background in computer architecture and memory/storage systems spans the stack, from devices and the OS up to serving frameworks.

Education

University of Illinois Urbana-Champaign 2022–2028 (expected)
Ph.D., Computer Science. Advisor: Prof. Minjia Zhang

University of Michigan, B.S.E., Computer Engineering (dual degree) 2020–2022

Shanghai Jiao Tong University, B.S.E., Electrical and Computer Engineering (dual degree) 2018–2022

Research Experience

SuperGuard, co-primary author *To appear in HPCA 2027*

- Identified dispatch-path interference on GH200: CPU-side memory contention (LLC pressure, DRAM queuing) slows host kernel dispatch and idles a GPU whose compute, HBM, and C2C are unsaturated.
- Built an MPAM-based feedback controller that partitions the Grace CPU's LLC, driven by dispatcher IPC: up to 2.1× end-to-end speedup (1.22× average), averaging 96% of interference-free performance.
- Validated on RAG serving and agentic pipelines (vLLM) and CPU-offloaded training (DeepSpeed).

Memory and storage systems, first or co-primary author *NeurIPS'25*, *HPCA'25*, *MICRO'23*

- **TeraIO:** Built lifetime-aware tensor offloading that extends GPU memory with SSDs via GPUDirect Storage for LLM training; 1.47× faster on average than ZeRO-Offload/Infinity (80.7% of ideal).
- **SkyByte:** Co-designed the host OS and SSD controller of a memory-semantic CXL-SSD; 6.11× faster than current CXL-SSDs with 23.08× less flash I/O traffic on average.
- **G10:** Built a unified GPU memory–storage architecture with compiler-scheduled tensor migrations; up to 1.75× faster than prior GPU memory solutions (90.3% of ideal).

Industry Experience

IBM Research, Yorktown Heights, NY May–Aug 2026
Research Intern, Hybrid Cloud. *Mentor: David Grove*

- Built an analytical cost model for IBM's Spyre dataflow AI accelerator, merged upstream into open-source torch-spyre, the PyTorch compiler backend, as a pass for comparing execution plans without running them (PR #3572).

T-Head Division, Alibaba Cloud, Hangzhou, China Jul–Aug 2021
Software R&D Intern (*Host: Yunhai Shang*). Optimized the JPEG library for RISC-V vector processors.

Skills

GPU & Kernels	CUDA (Graphs, Virtual Memory Management, TMA), CuTe DSL, Nsight / CUPTI
ML Systems	vLLM internals, FlashInfer, PyTorch, DeepSpeed; speculative decoding, MoE expert parallelism and offloading, KV-cache management
Memory & I/O	CXL, GPUDirect Storage, NVLink-C2C, InfiniBand, UVM / ATS, MPAM / resctrl
Architecture	NVIDIA GH200 / H200; gem5, Accel-Sim (GPGPU-Sim), MacSim, DRAMSim2
Languages	Python, C/C++, Verilog/SystemVerilog

Publications [G]

* co-primary authors

Under Review

1 paper under review (ML systems; details on request).

Conference and Journal Papers

[1] Hsu-Tzu Ting*, **Haoyang Zhang***, Xinyu Lian, and Minjia Zhang. "**Where Did the GPU Time Go? Uncovering and Controlling Hidden Interference in LLM Serving and Training on Superchips**". (To appear in **HPCA** 2027)

[2] Ziqi Yuan*, **Haoyang Zhang***, Yirui Eric Zhou*, Apoorve Mohan, I-Hsin Chung, Seetharami Seelam, and Jian Huang. "**TeraIO: Cost-Efficient LLM Training with Lifetime-Aware Tensor Offloading via GPUDirect Storage**". In: *Conference on Neural Information Processing Systems* **(NeurIPS)**. 2025.

[3] **Haoyang Zhang***, Yuqi Xue*, Yirui Eric Zhou, Shaobo Li, and Jian Huang. "**SkyByte: Architecting an Efficient Memory-Semantic CXL-based SSD with OS and Hardware Co-design**". In: *IEEE International Symposium on High-Performance Computer Architecture* **(HPCA)**. 2025.

[4] **Haoyang Zhang***, Yirui Eric Zhou*, Yuqi Xue, Yiqi Liu, and Jian Huang. "**G10: Enabling An Efficient Unified GPU Memory and Storage Architecture with Smart Tensor Migrations**". In: *IEEE/ACM International Symposium on Microarchitecture* **(MICRO)**. 2023.

[5] Jiacheng Ma, Gefei Zuo, Kevin Loughlin, **Haoyang Zhang**, Andrew Quinn, and Baris Kasikci. "**Debugging in the Brave New World of Reconfigurable Hardware**". In: *ACM International Conference on Architectural Support for Programming Languages and Operating Systems* **(ASPLOS)**. 2022.

[6] Xingyue Qian, Jian Shi, Li Shi, **Haoyang Zhang**, Lijian Bian, and Weikang Qian. "**Scheduling Information-Guided Efficient High-Level Synthesis Design Space Exploration**". In: *IEEE International Conference on Computer Design* **(ICCD)**. 2022.

[7] Chen Chen, **Haoyang Zhang**, Kaicheng Guo, Xingzi Yu, Weidong Qiu, Zhengwei Qi, and Haibing Guan. "**Exploring Efficient Hardware Accelerator for Learning-Based Image Compression**". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* **(TCAD)** 44.6 (2025).

Honors, Teaching & Service

MICRO 2023 Travel Grant — 2023

Roger King Scholarship, University Honors, and Dean's List, University of Michigan — 2020–2021

Graduate TA, Computer Architecture (CS 233), UIUC — FA 2024, SP 2025, FA 2025

TA, Computer Architecture (VE 470) and Honors Physics (VP 160), SJTU — SU 2022, SU 2020

Artifact Evaluation Committee, HPCA — 2025